Radxa Cubie A5E & 7semi 4G LTE Guide

Complete Step-by-Step Setup, AT Commands Reference, and Remote SSH Configuration

Board: Radxa Cubie A5E

Modem: 7semi 4G LTE (USB RNDIS/ECM)

Carrier Tested: Jio LTE (India)

1 Initial Hardware & Tool Setup

Before beginning software configuration, assemble hardware and install essential modem tools on the Radxa board.

1. Hardware Assembly:

- Insert an active nano-SIM card into the 7semi 4G LTE board.
- Connect 4G LTE antennas securely to the SMA/IPEX ports.
- Connect the 7semi module to a USB 2.0/3.0 port on the Radxa Cubie A5E.
- Connect Radxa Cubie A5E to your local router via Ethernet (`eth1`).

2. Install Required Packages:

```
sudo apt update
sudo apt install -y modemmanager usb-modeswitch minicom network-manager
```

2 Interface Verification & Diagnostics

Verify that the Linux kernel detects the USB module and serial interfaces:

```
# Check active network devices
nmcli device

# Check serial interface nodes created by the modem
ls -l /dev/ttyUSB*
```

Standard detection output:

- **Network Interface:** `enxae0c29a39b6d` (USB Ethernet / RNDIS mode)
- **AT Command Port:** `/dev/ttyUSB2`

3 Modem Configuration via AT Commands (Minicom)

Access the AT control port using `minicom` to configure APN and packet routing:

```
sudo minicom -D /dev/ttyUSB2 -b 115200
```

Minicom Navigation Tip

To exit minicom: press **Ctrl + A**, release, then press **X**, and hit **Enter**.

Execute these AT commands inside minicom:

AT COMMAND	EXPECTED OUTPUT	DESCRIPTION
`AT`	OK	Verifies serial communication
`AT+CPIN?`	+CPIN: READY	Checks SIM card detection
`AT+CSQ`	+CSQ: 30,99	Checks signal strength (Target: 10–31)
`AT+CREG?`	+CREG: 0,1	Checks network registration (Home network)
`AT+CGDCONT=1,"IP","jionet"`	OK	Sets APN (Use "jionet" for Jio, "airtelgprs.com" for Airtel)
`AT+CGATT=1`	OK	Attaches to LTE packet service
`AT+CGACT=1,1`	OK	Activates PDP context 1 for data flow
`AT+CGPADDR=1`	+CGPADDR: 1,"10.x.x.x"	Confirms active 4G IP address assigned by tower
`AT+QCFG="nat",1`	OK	Enables internal NAT packet routing on modem
`AT+QCFG="usbnet",1`	OK	Sets USB network mode to ECM/Ethernet
`AT+CFUN=1,1`	OK	Reboots LTE modem firmware to apply changes

4 NetworkManager Routing & DNS Configuration

After modem reboot, apply custom DNS settings and restart the connection in Linux:

```
# Set public DNS servers on the 4G network connection
sudo nmcli connection modify "Wired connection 3" ipv4.dns "8.8.8.8 1.1.1.1"

# Restart the connection interface
sudo nmcli connection down "Wired connection 3"
sudo nmcli connection up "Wired connection 3"

# Restart NetworkManager daemon
sudo systemctl restart NetworkManager
```

Verify Internet Connection:

```
# Ping modem gateway
ping -c 4 192.168.43.1

# Ping Google DNS
ping -c 4 8.8.8.8

# Ping domain name
ping -c 4 google.com
```

5 Remote SSH Setup over 4G LTE (Tailscale)

Carrier-Grade NAT (CGNAT) Notice

Jio and most 4G mobile networks do not issue public IP addresses to SIM cards. Direct SSH into the SIM's IP is blocked by CGNAT. Using Tailscale provides a secure, direct VPN tunnel to your Radxa board from anywhere.

1. Enable SSH Service on Radxa:

```
sudo systemctl enable --now ssh
```

2. Install and Authenticate Tailscale:

```
curl -fsSL https://tailscale.com/install.sh | sh
sudo tailscale up
```

Follow the login URL printed in the terminal to connect the Radxa Cubie A5E to your Tailscale account.

3. Connect Remotely via SSH:

On your remote PC, laptop, or phone (with Tailscale installed and active):

```
ssh radxa@<TAILSCALE_IP_ADDRESS>
```

Setup Complete!

Your Radxa Cubie A5E is now fully configured with auto-connecting 4G LTE internet and secure remote SSH access from any network globally.